

Project Initialization and Planning Phase

Date	1 October 2026
Team ID	SWTID-2026-3594
Project Title	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to secure institution records in ServiceNow using script-controlled Access Control Lists (ACLs), improving data protection and accuracy of access. It tackles over-exposure of records, promising tighter security, clearer role separation and more confident users. Key features include a script-controlled READ ACL with a Branch condition and separate CREATE, WRITE and DELETE ACLs.

Project Overview	
Objective	To restrict users from viewing records unless they meet specific conditions, so that only users of the EEE branch can view records whose Branch field is EEE while administrators retain full access.
Scope	Creation of the EEE User and custom roles bb1 - bb4, the custom table u_institution_details, and READ, CREATE, WRITE and DELETE ACLs with verification by impersonation.
Problem Statement	
Description	Records in the Institution Details table are visible to every user with table access, and the rights to create, edit and delete are not separated by role.
Impact	Solving this protects student data, reduces the risk of unauthorized changes, and makes it clear which role is needed for each action.
Proposed Solution	
Approach	Use script-controlled ACLs in ServiceNow: a READ ACL with the data condition Branch is EEE, role bb1 and an admin-bypass script, plus one ACL per operation (bb2, bb3, bb4).
Key Features	- Script-controlled READ ACL with a Branch is EEE data condition. - Role-separated CREATE, WRITE and DELETE ACLs (bb2, bb3, bb4). - Admin bypass so administrators can view all records. - Verification by impersonating the EEE User and the Admin.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Hosted by ServiceNow (PDI); any PC with a modern browser
Memory	RAM specifications	4 GB or more on the client device
Storage	Disk space for data, models, and logs	Managed by the ServiceNow instance; no local storage needed

Resource Type	Description	Specification/Allocation
Software		
Frameworks	Platform frameworks	ServiceNow platform (Users, Roles, Tables, ACLs)
Libraries	Additional libraries	Glide server-side API (GlideSystem: gs.hasRole)
Development Environment	IDE	ServiceNow browser-based script editor (Chrome / Edge)
Data		
Data	Source, size, format	Sample records entered manually in u_institution_details for ECE, EEE and CSE